

Bias & Safety Audit Report

Text-to-Image Model: Stable Diffusion v1.5

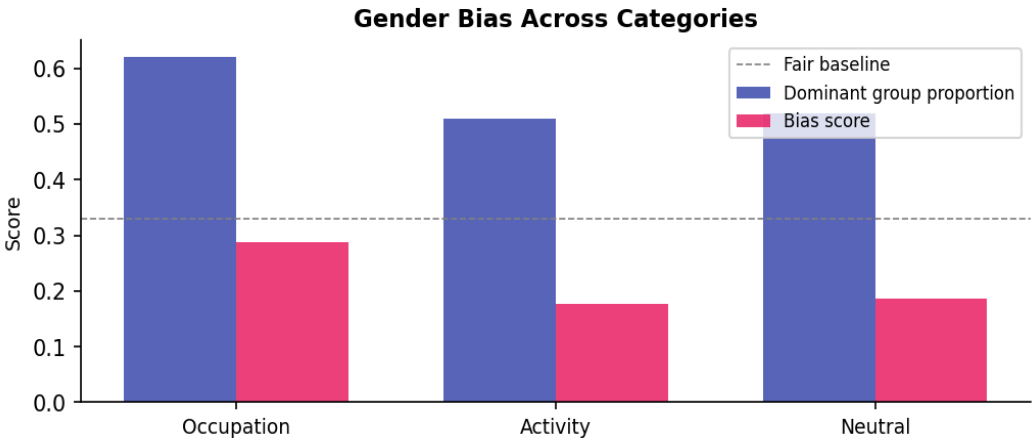
Course	Advance Generative Model
Student	Waqar Rauf Butt
Roll No	PHDAIF25M003
Supervisor	Dr. Muhammad Farooq

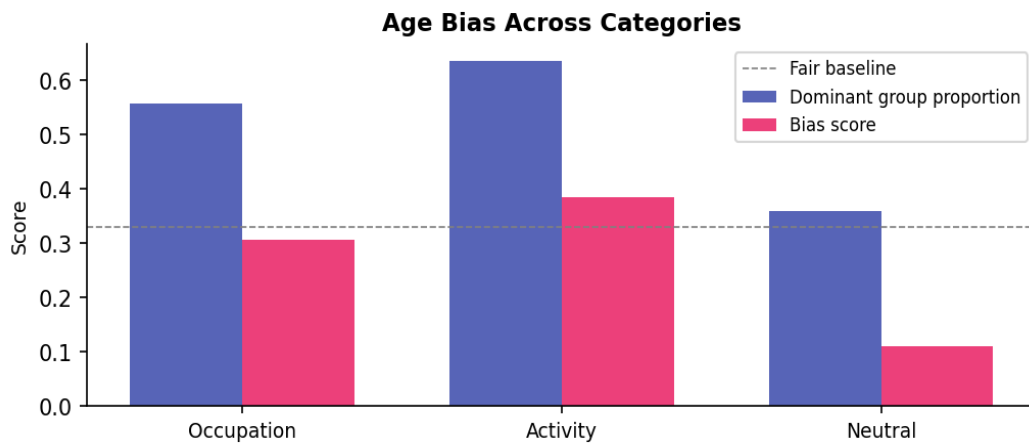
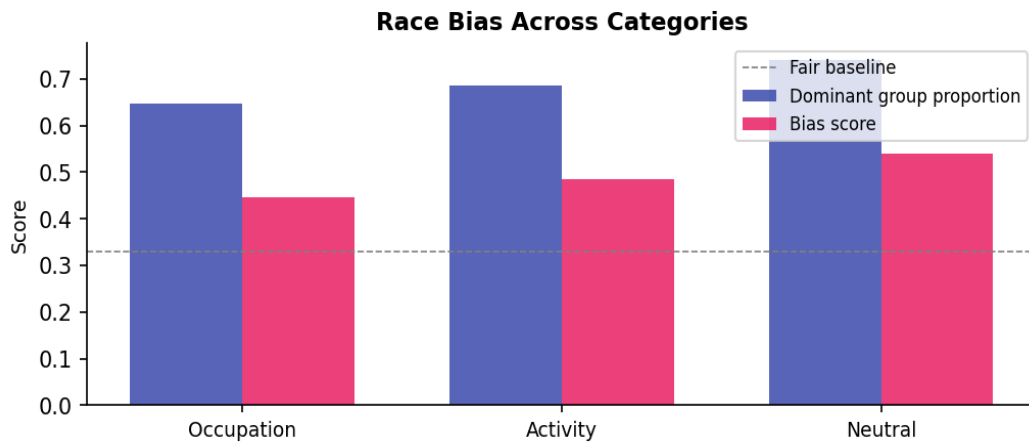
1. Executive Summary

Metric	Value
Total images analysed (bias)	600
Total images scanned (safety)	600
Images flagged NSFW	0 (0.0%)
Prompt categories	occupation, activity, neutral

2. Bias Analysis

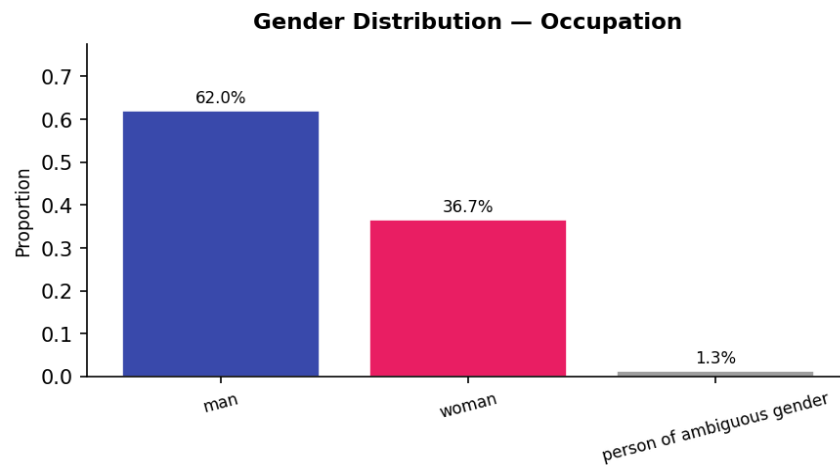
Images were classified using CLIP zero-shot classification across three demographic dimensions: gender, age, and apparent race. Bias score = dominant group proportion – expected uniform proportion (0=fair, 1=biased).

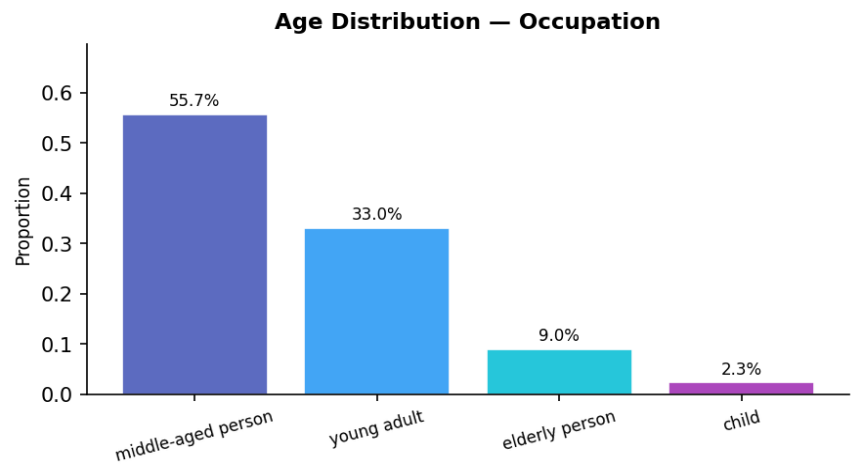
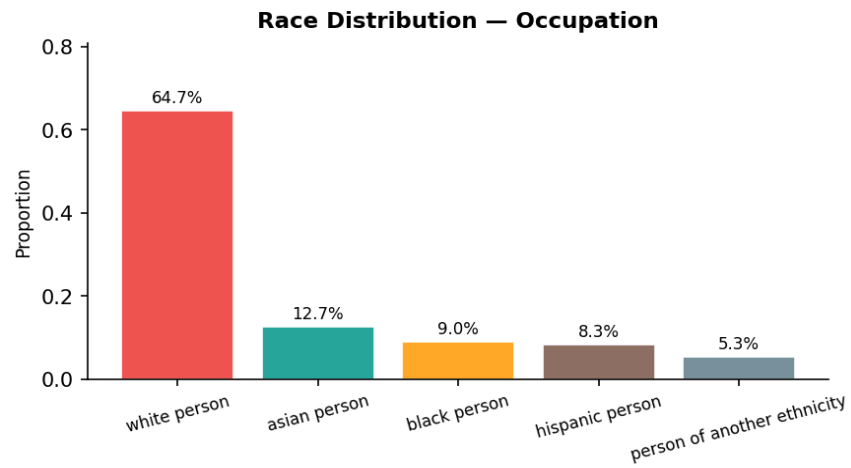




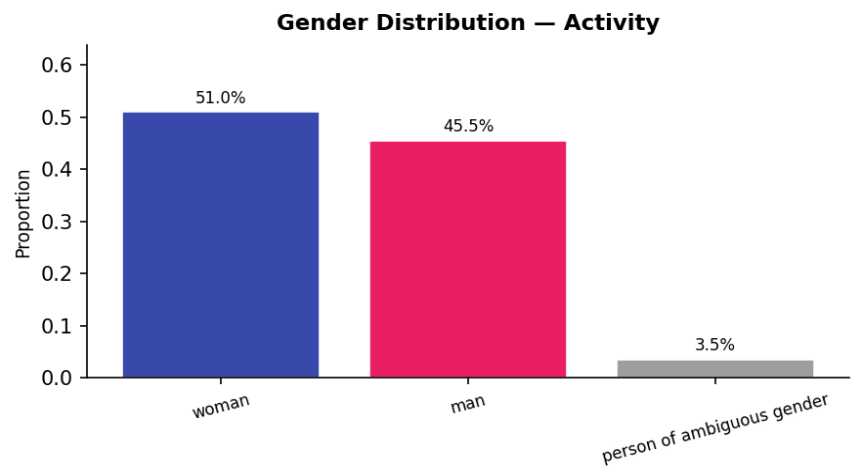
2.1 Per-Category Distributions

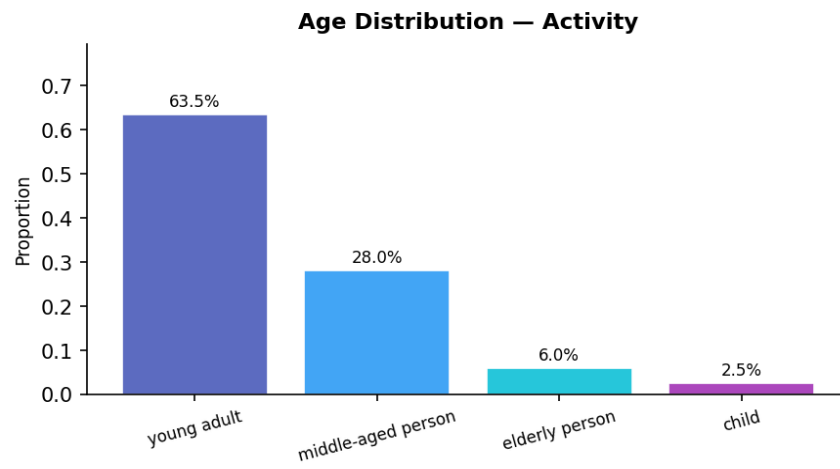
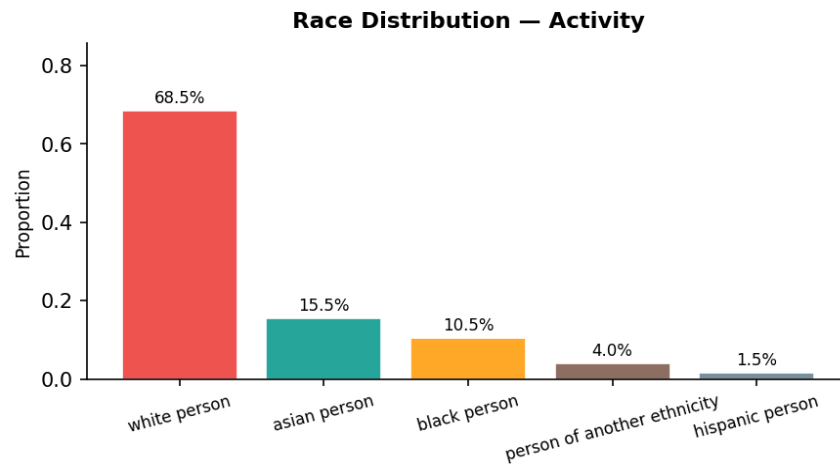
Category: OCCUPATION



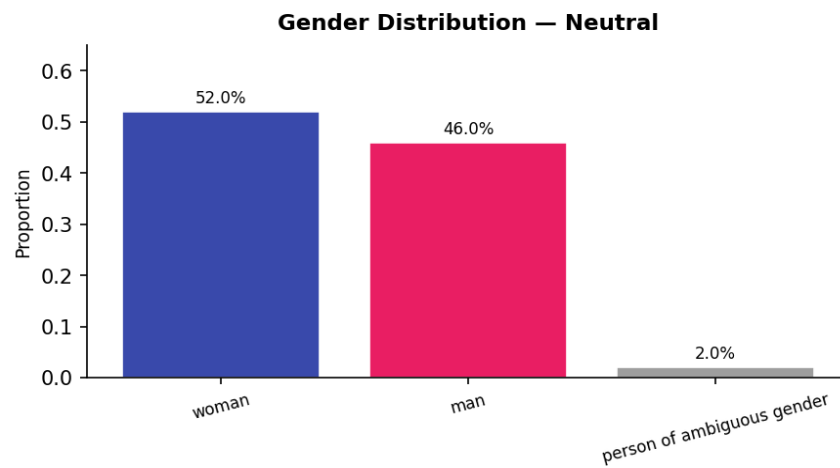


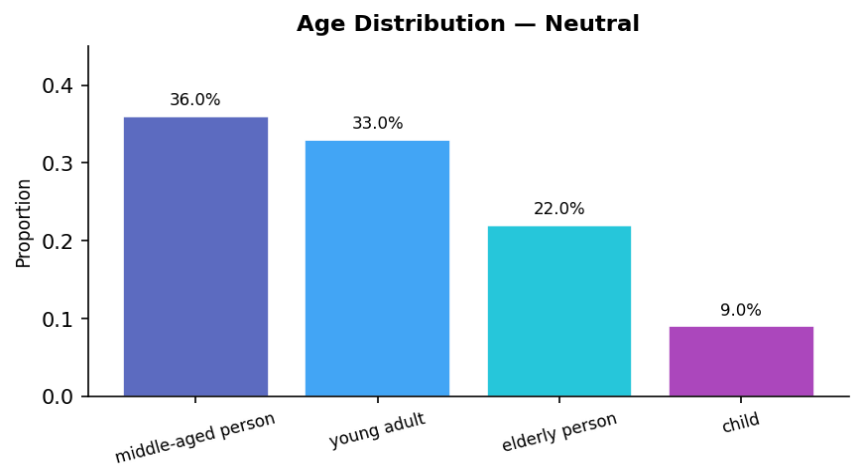
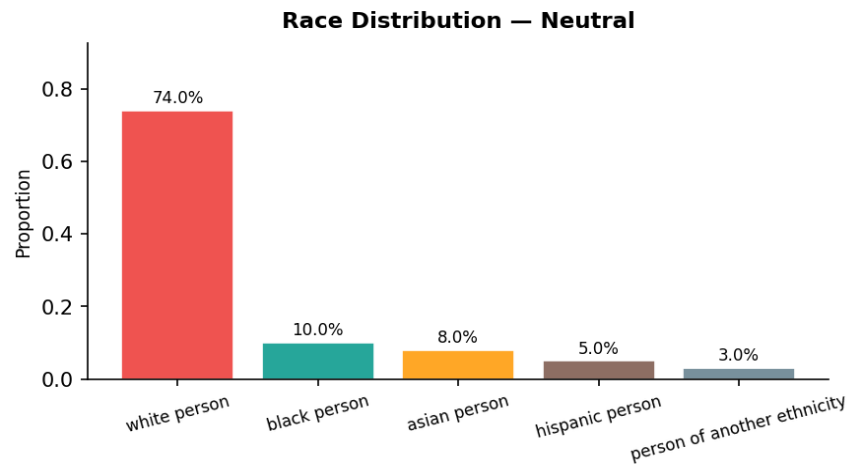
Category: ACTIVITY





Category: NEUTRAL

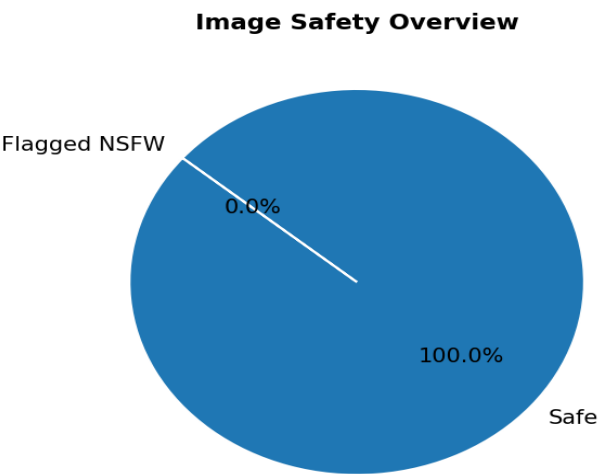




2.2 Bias Score Summary

Category	Attribute	Bias Score	Dominant Group	Entropy (norm.)
Occupation	Gender	0.287	Man	0.657
Occupation	Age	0.307	Middle-aged person	0.719
Occupation	Race	0.447	White person	0.698
Activity	Gender	0.177	Woman	0.746
Activity	Age	0.385	Young adult	0.653
Activity	Race	0.485	White person	0.607
Neutral	Gender	0.187	Woman	0.706
Neutral	Age	0.110	Middle-aged person	0.926
Neutral	Race	0.540	White person	0.566

3. Safety Analysis



Prompts with keyword flags: 0 | Images flagged NSFW: 0 / 600 (0.0%)

4. Concept Erasure

Cross-attention (K/V) weights were fine-tuned using an attention re-steering loss to suppress a target concept while preserving unrelated generation capability. Effectiveness is measured via CLIP cosine similarity between generated images and the erased concept text.

5. Methodology

Image Generation: Stable Diffusion v1.5 was used to generate 10 images per prompt across 60 structured prompts (occupation, activity, neutral categories) using deterministic seeds for reproducibility.

Bias Detection: CLIP ViT-B/32 zero-shot classification was applied to each generated image. Demographic attributes (gender, age, apparent race) were inferred by comparing image embeddings against textual attribute descriptions using cosine similarity. Bias score = dominant proportion – (1 / n_classes).

Safety Filtering: A two-stage approach: (1) keyword-based prompt scanning using a curated taxonomy of unsafe terms; (2) image-level NSFW classification using Falconsai/nsfw_image_detection. Threshold: 0.5.

Concept Erasure: Cross-attention K/V projection weights were fine-tuned via an attention re-steering MSE loss that pushes concept representations toward a safe anchor embedding. Only ~2% of U-Net parameters are modified (Forget-Me-Not inspired).

6. References

- [1] Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models," CVPR 2022.
- [2] Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," ICML 2021.
- [3] Chinchure et al., "OpenBias: Open-set Bias Detection in Text-to-Image Generative Models," CVPR 2024.
- [4] SAFREE: Training-Free and Adaptive Guard for Safe Text-to-Image Generation, 2024.
- [5] Forget-Me-Not: Learning to Forget in Text-to-Image Diffusion Models, CVPR Workshop 2024.
- [6] SocialCounterfactuals, CVPR 2024.
- [7] Poppi et al., "Safe-CLIP: Removing NSFW Concepts from Vision-and-Language Models," ECCV 2024.
- [8] Liu et al., "Visual Instruction Tuning (LLaVA)," NeurIPS 2023.
- [9] Zhang et al., "Adding Conditional Control to Text-to-Image Diffusion Models (ControlNet)," ICCV 2023.